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Roll No.....

Degree: B.Tech. Semester: III - Course work  
MID-SEMESTER EXAMINATION, SEPTEMBER 2025  
Course Title: Mathematics for Communication and Signal Processing  
Course Code: ECMTC301

Time: 1:30 Hours

Max. Marks: 20

Note: Attempt all questions in given order only. Missing data/ information (if any), maybe suitably assumed and mentioned in the answer.

| Q. No. | Question                                                                                                                                                                                                                                                               | Marks | CO  |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|
| 1a     | Prove that the set $H = \left\{ A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_2(\mathbb{R}) \mid a + 2b - c = d \right\}$ forms a subgroup of $M_2(\mathbb{R})$ , where $M_2(\mathbb{R})$ denotes the group of all $2 \times 2$ real matrices under addition. | 2     | CO1 |
| 1b     | Define cyclic group. Is $U(16)$ a cyclic group?                                                                                                                                                                                                                        | 2     | CO1 |
| 2a     | Check whether the ring $2\mathbb{Z}$ is isomorphic to the ring $7\mathbb{Z}$ . Justify your answer.                                                                                                                                                                    | 2     | CO1 |
| 2b     | Consider the ring $\mathbb{Z}_6 = \mathbb{Z}/6\mathbb{Z}$ .<br>(a) Check whether $\mathbb{Z}_6$ has any zero divisors.<br>(b) Is $\mathbb{Z}_6$ a field. Justify.                                                                                                      | 2     | CO1 |
| 3a     | Determine whether the set $W = \{(x, y, z) \in \mathbb{R}^3 \mid x - y = 2z\}$ forms a subspace of $\mathbb{R}^3$ ?                                                                                                                                                    | 2     | CO2 |
| 3b     | Consider the vector space $\mathbb{P}_2(\mathbb{R})$ of polynomials of degree at most 2 over real numbers. Is the set $\{1 + x, x + x^2, 1 + x^2\}$ forms a basis for the vector space $\mathbb{P}_2$ ? Justify your answer.                                           | 2     | CO2 |
| 4a     | Let $U = \{(x, y, z, w) \in \mathbb{R}^4 : x + 2y - z = 0, 3x - y + w = 0\}$ . Find a basis for $U$ and hence determine the dimension of $U$ .                                                                                                                         | 2     | CO2 |
| 4b     | Define $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ by $T(a_1, a_2, a_3) = (2a_1 + a_2 - a_3, a_3 - 2a_1)$ . Show that $T$ is a linear transformation.                                                                                                                  | 2     | CO2 |
| 5a     | Using rank nullity theorem show that the linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^4$ given by $T(x, y, z) = (x - 3y, y - z, 0, z - 2x)$ is not bijective.                                                                                        | 2     | CO2 |
| 5b     | For the vector field $\mathbf{F}(x, y, z) = (yz)\mathbf{i} + (xz)\mathbf{j} + (xy)\mathbf{k}$ : (i) Compute the curl $\mathbf{F}$ i.e. $\nabla \times \mathbf{F}$ . (ii) Compute the divergence of the curl, $\nabla \cdot (\nabla \times \mathbf{F})$ .               | 2     | CO3 |